

Exp 8

Date: 1/9/21

Roll No: 7

Compute Parameters like RANGE,
VELOCITY, Power using RADAR Equation.

AIM:

To Compute the maximum detection range
and received power of a RADAR system
using the RADAR Range operation.

APPARATUS REQUIRED:

- * Laptop with i3 processor
- * ~~SCI~~ Lab software.

CODING:

clc;

clear;

$$P_t = 1000 \text{ W}$$

$$G = 31 \text{ dB}$$

$$G = 10^{(G - \text{dB}/10)}$$

$$f = 100000000 \text{ Hz}$$

$$c = 300000000 \text{ m/s}$$

$$\begin{aligned} G &= 1 \\ P_t &= 1000 \\ P_r &= 1.445 \times 10^{-11} \end{aligned}$$

S.NO	Description	Theoretical	Practical
1.	R vs Pt Pt = 1 : 10 : 10000 (Pt = 9.83 e°dB) Gt = 31 Gr = 31 λ = 0.03 m σ = 1 m² PR = 10⁻¹² W Pt = 8.697 W	$R = \left[\frac{P_r G_t G_r \lambda^2 \sigma}{(4\pi)^5 P_t} \right]^{1/4}$ $R = \left[\frac{8.697 \times 30 \times 30 \times (0.03)^2 \times 1}{(4\pi)^3 \times 10^{-12}} \right]^{1/4}$ $= \left[\frac{7.04757}{1.9844 \times 10^{-9}} \right]^{1/4}$ $= 244.1 \text{ m}$	R = 247.1 m
2.	R vs Pr Pr = -15 : 1 : 9 (1084 dB) Gt = 30 Gr = 30 λ = 0.03 m σ = 1 Pt = 1000 Gt = 1.445 × 10¹¹	$R = \left[\frac{1000 \times 30 \times 30 \times (0.03)^2 \times 1}{(4\pi)^2 \times 1.99526 \times 10^{11}} \right]^{1/4}$ $= \left[\frac{810}{3.9594 \times 10^{-8}} \right]^{1/4}$ $= \left[\frac{810}{3.9594 \times 10^{-8}} \right]^{1/4}$ $= 378.19$	R = 398.19

$$\lambda = c/f;$$

$$\sigma = 10;$$

$$L - \text{dB} = 3;$$

$$L = 10^{(L - \text{dB}/10)};$$

$$S_{\min} = 1e-13;$$

$$A_{\max} = \left((P_t * G^2 * \lambda^2 * \sigma) / \left((4 * \pi)^3 * S_{\min} * L \right) \right)^{1/4};$$

disp (" = = = = = ");

disp ("..... RADAR SIMULATION RESULT.....");

disp (" = = = = = ");

disp (" Transmitted power (w): ");

disp ("Pt");

disp (G - dB);

disp (* frequency (GHz): ");

disp (f/1e9);

disp ("wavelength (m): ");

disp (~~lambda~~);

disp (sigma);

disp (" system loss (dB): ");

disp (L - dB);

disp (" minimum detectable: ");

disp (S_min);

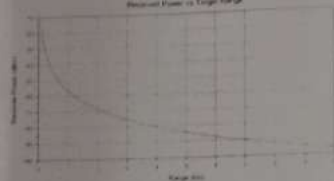
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disp ("Maximum Radar Range (km) :");
disp (Rmax (1000));
Pr = 100 : 100 : 10000 ;
Pr = (Pt * G ^ 2 * lambda ^ 2 * sigma) ./
      ((4 * %pi) ^ 3 * R.^4 * L);
Pr - dBm = 10 * log10 (Pr / 1e - 3);
Pt - range = 100 : 100 : 10000 ;
Rmax - Pt = ((Pt - range * G ^ 2 * lambda ^ 2 * sigma) ./
              ((4 * %pi) ^ 3 * sin ^ 2 L)) . ^ (1/4);
Rmax - Pt - Km = Rmax - Pt / 1000 ;
G - dB - range = 10 : 1 : 40 ;
G - range = 10 . ^ (G - dB - range / 10) ;
Rmax - G - Km = Rmax - G / 1000 ;
Scf (1) ;
Subplot (2, 2, 1);
plot (R/1000, Pr - dBm);
xlabel ("Range (Km)");
ylabel ("Received Power (dBm)");
title ("Received Power Vs Target Range");
xgrid ();
Subplot (2, 2, 2);
plot (Pt - range / 1000, Rmax - Pt - Km);
xlabel ("Transmitted power (KW)");
ylabel ("Maximum Range (Km)");
title ("Maximum Range Vs Transmitted power");
xgrid ();

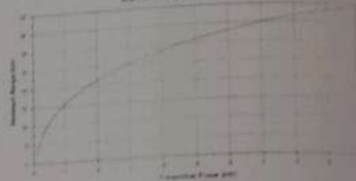
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1. ☐ Plot
 2. ☐ Table
 3. ☐ V & P
 4. ☐ P & R

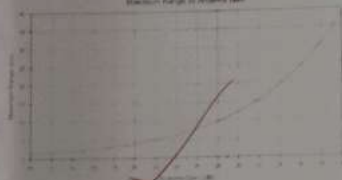
Maximum Power vs Target Range



Maximum Range vs Transmitted Power



Maximum Range vs Antenna Gain



```

subplot (2, 2, 3);
plot (G - dB - range, R_max - G - Km);
x_label ("Antenna Gain (dB)");
y_label ("maximum Range (Km)");
title ("maximum Range Vs Antenna Gain");
x_grid ();

```

Marks allocation :

Scilab program numerical implementation (20)	Wave form and labulation (20)	Calculation and result (20)	viva - voice (20)	Result Observation (20)	Total marks
20	20	20	20	16	96

RESULT:

Thus, the maximum range of radar system
~~calculated~~ using radar equation is successfully
 verified using scilab programming